

DNA Sequence Analysis – Assignment Script

Identifying Information

Programmer: Puja

Language: R (Version 4.3.0 or higher recommended)

Date Submitted: 09 November 2025

Script Filename: `assignment.R`

Data File: `chr1_GL383518v1_alt.fa`

Purpose of the Script

This R script performs a comprehensive analysis of a DNA sequence stored in a FASTA file. It is divided into four main parts:

- Part 1:**
Reads the DNA sequence from the FASTA file and prints the 10th and 758th nucleotide bases.
- Part 2:**
Generates the reverse complement of the sequence and extracts specific nucleotide positions.
- Part 3:**
Divides the sequence into kilobase (1,000 base) segments and counts occurrences of each nucleotide (A, C, G, T) per segment.
Includes data validation and QC (Quality Control) checks.
- Part 4:**
Converts the nucleotide count data into a data frame, calculates total bases per kilobase, and verifies expected counts.
Includes explanations for discrepancies due to partial kilobases.

Each section includes **QC checkpoints** to validate input files, confirm data integrity, and ensure correct processing.

Files Required

File Name	Description
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<code>chr1_GL383518v1_alt.fa</code>	Input FASTA file containing the DNA sequence of chromosome 1 alternative assembly (required).
<code>assignment.R</code>	Main R script containing all four parts of the analysis.

Libraries and Packages Required

Before running the script, make sure the following R packages are installed:

Package Purpose

Package	Purpose
<code>stringr</code>	For string manipulation and counting specific nucleotides.
<code>dplyr</code>	For data frame manipulation and summarization.

Instructions to Run the Script

Follow these steps to execute the program successfully:

- Set up working directory**
 - Place the FASTA file (`chr1_GL383518v1_alt.fa`) and the R script (`assignment.R`) in the same folder.
- Run the script**

Execute the script in R using:

```
source("assignment.R")
```

- QC Checks**
 - The script includes built-in QC checkpoints that will:
 - Confirm file presence.
 - Validate sequence length.
 - Verify structure of generated lists and data frames.
 - Identify any missing or unexpected nucleotide counts.
- Expected Output**
 - Printed details of the 10th and 758th bases.
 - The 79th base and 500–800 range from the reverse complement.

- A list of nucleotide counts per kilobase.
- A data frame summarizing nucleotide distribution across all kilobases.
- QC summary messages and explanations of observed differences.

Output Files Generated

Output Type	Description
Console Output	Displays QC messages, base positions, and summary tables.
Data Frame (df)	Created in memory containing nucleotide counts and total base sums for each kilobase.
QC Logs	Printed in the R console to confirm successful processing and highlight discrepancies.

Notes

- The final kilobase in the DNA sequence may contain fewer than 1,000 bases, leading to sums less than the expected total.
- This behavior is **biologically and computationally normal** since DNA sequence lengths are not always perfect multiples of 1,000.